

Track 6: Cybersecurity & Ethical Hacking

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MR ROKESH TECHNOLOGY — Training & Internship Programs

Skill path: Fundamentals → Networking → Recon → Exploitation → Web Security → Reporting

Related: [Master Syllabus](#) | [Duration Framework](#) | [Project Progression](#)

Track Overview

Who this is for: Aspiring security analysts, penetration testers, and SOC professionals.

Prerequisites: Basic networking and Linux helpful; taught from fundamentals in 1-month+ tiers.

Tools & lab environment: Kali Linux (VM), VirtualBox/VMware, Metasploitable, DVWA, OWASP Juice Shop, Nmap, Burp Suite Community, Wireshark, TryHackMe/HackTheBox (optional).

Legal/ethical framework: All labs in isolated VMs only; written authorization required for any external testing; CEH code of ethics reviewed Week 1.

Certification targets (6–9 month): CEH or CompTIA Security+ prep

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Security awareness	Vulnerability assessment report
1 Month	80–100	Junior analyst basics	Mini pentest report
3 Months	250–300	Internship-ready	Full pentest report on test app
6 Months	500–600	Pentester entry-level	End-to-end pentest + professional report
9 Months	750–900	Red/blue team ready	Enterprise pentest + IR simulation

Weekly Rhythm (All Durations)

Mon–Thu lab exercises | Fri lab report | Sat CTF/review | Sun document findings (Markdown/PDF)

2-Week Crash Course

Mini Project: Vulnerability assessment report | **Level:** L1

Days	Topics	Lab	Self-Check
D1–2	Cyber fundamentals, CIA triad, legal/ethical framework	Security awareness quiz	Define confidentiality, integrity, availability
D3–4	Networking (TCP/IP), Linux basics, Kali setup	Kali lab setup	Explain OSI model layers 3–7
D5–6	Reconnaissance, Nmap scanning, footprinting	Scan Metasploitable VM	Nmap scan types (-sS, -sV)
D7–8	Web security: OWASP Top 10 (SQLi, XSS basics)	DVWA exploitation labs	Manual vs automated SQLi
D9–10	Password security, basic reporting	Vulnerability assessment report	CVSS scoring basics

Submission checklist: - ☐ Kali VM configured with snapshot - ☐ Nmap scan output attached - ☐ DVWA SQLi and XSS screenshots - ☐ 3-page assessment report with remediation recommendations

1-Month Foundation

Capstone: Mini pentest report | **Level:** L2

Week	Topics	Lab/Project
W1	Networking, Linux, Kali, cryptography basics	Lab setup + encryption lab
W2	Recon, scanning (Nmap), enumeration	Footprinting report
W3	System hacking, web app security (Burp Suite)	DVWA + OWASP Juice Shop
W4	Wireless basics, incident response, pentest report writing	Mini pentest report

W4 acceptance criteria: - ☐ Executive summary + technical findings - ☐ Minimum 5 vulnerabilities documented with severity - ☐ Remediation steps for each finding - ☐ Methodology section (recon → scan → exploit → report)

3-Month Job Ready

Capstone: Full pentest report on test app | **Level:** L3

Month	Topics	Projects
M1	Network security, Linux admin, cryptography	5 guided labs
M2	Recon, scanning, enumeration, exploitation (Metasploit)	CTF challenges (10)
M3	Web app pentesting (OWASP Top 10), reporting, capstone	Full pentest report on test app

M3 Capstone acceptance criteria: - ☐ All OWASP Top 10 categories tested (where applicable) - ☐ Burp Suite project file submitted - ☐ Professional report format (15+ pages) - ☐ Retest verification for critical findings

6-Month Professional

TCM Security + Texas A&M syllabus aligned

Month	Topics	Labs
M1	Cyber fundamentals, networking, Linux, cryptography	Labs 1–3
M2	Footprinting, scanning, enumeration, system hacking	Labs 4–6
M3	Web app security (SQLi, XSS, CSRF, IDOR), Burp Suite	Labs 7–9
M4	Wireless security, malware analysis basics, Active Directory attacks	AD lab build
M5	Cloud security basics, SIEM intro, incident response	IR simulation
M6	Full penetration testing methodology, CTF, cert prep	End-to-end pentest + professional report

M6 Capstone acceptance criteria (L4): - [] AD lab exploited with documented attack path - [] Cloud misconfiguration findings (S3/public bucket demo) - [] SIEM alert triage simulation completed - [] 25+ page report with executive and technical audiences

9-Month Advanced

Capstones: Enterprise network pentest + Blue team IR simulation | **Level:** L5

Additional modules: Advanced AD exploitation, red team vs blue team exercises, SOC analyst workflows, digital forensics basics, bug bounty methodology.

Assessment Rubric (Track-Specific)

Component	Weight	Criteria
Weekly Quiz	20%	Concepts, tools, legal/ethical scenarios
Lab Completion	40%	Lab screenshots, commands logged, VMs configured
Project	30%	Report quality, finding accuracy, remediation
Portfolio	10%	GitHub with sanitized lab notes; TryHackMe profile

Recommended Resources

- [TCM Security – Practical Ethical Hacking](#)
- [OWASP Top 10](#)
- [TryHackMe](#) / [HackTheBox](#)

Appendix: Mentor Notes

Common pitfalls: Running scans against unauthorized targets; skipping report writing; tool dependency without understanding underlying vulnerability.

Hardware: 16 GB RAM recommended for VMs; SSD required; never use production network for attack labs.

Cross-track: [Track 7 — DevOps](#) for DevSecOps; [Track 8 — Cloud](#) for cloud security labs.